STUDENT ANALYSIS

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LOAD THE NECESSARY PACKAGE

```r
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library("lubridate")
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

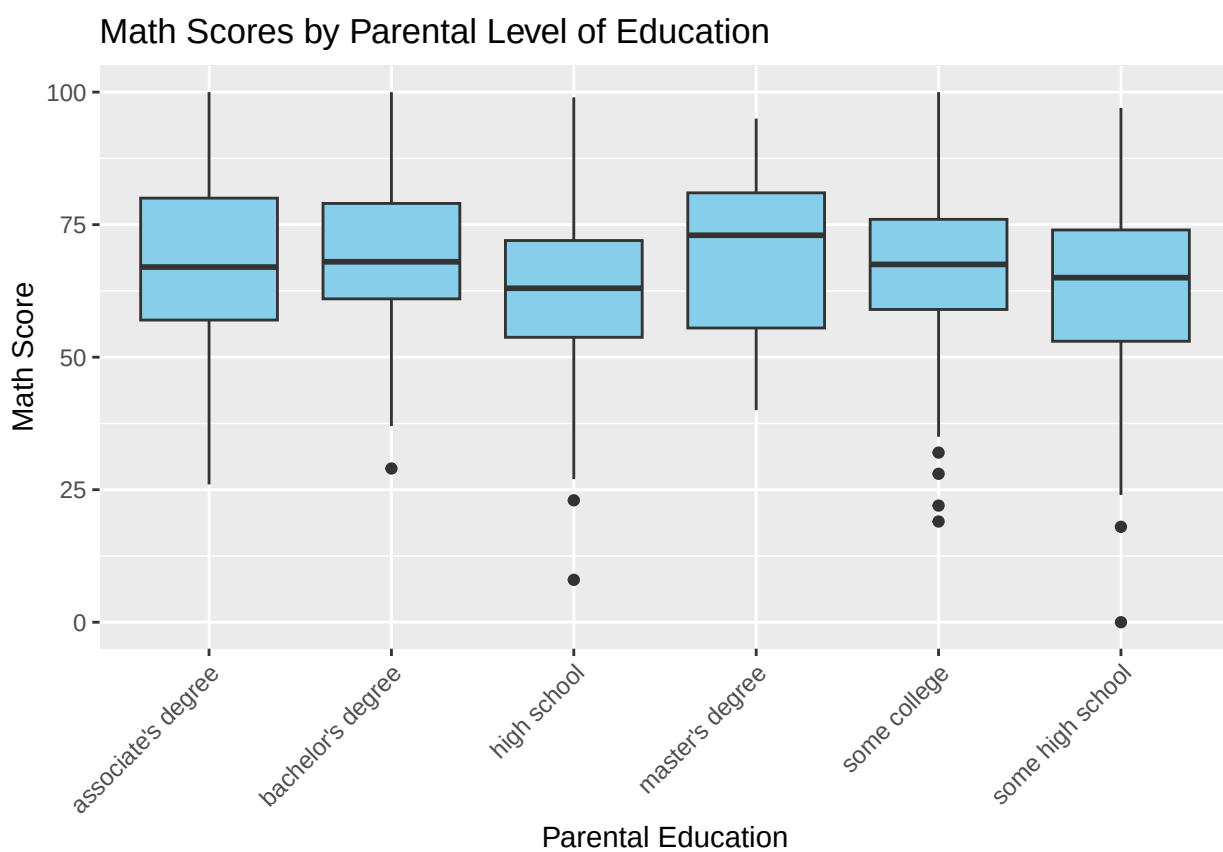
```r
library("ggplot2")
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats 1.0.1     v stringr 1.6.0
## v purrr   1.2.1     v tibble  3.3.1
## v readr   2.2.0     v tidyr   1.3.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(tidyverse)

# Load the dataset
df <- read.csv("StudentsPerformance.csv")

# Clean column names for easier coding (replaces spaces and slashes with underscores)
colnames(df) <- gsub("\\.", "_", make.names(colnames(df)))
# Resulting names: gender, race_ethnicity, parental_level_of_education, lunch,
# test_preparation_course, math_score, reading_score, writing_score
```

##QUESTION 1.MATH SCORES BY PARENTAL EDUCATION

```r
ggplot(df, aes(x = parental_level_of_education, y = math_score)) +
  geom_boxplot(fill = "skyblue") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Math Scores by Parental Level of Education",
       x = "Parental Education", y = "Math Score")
```



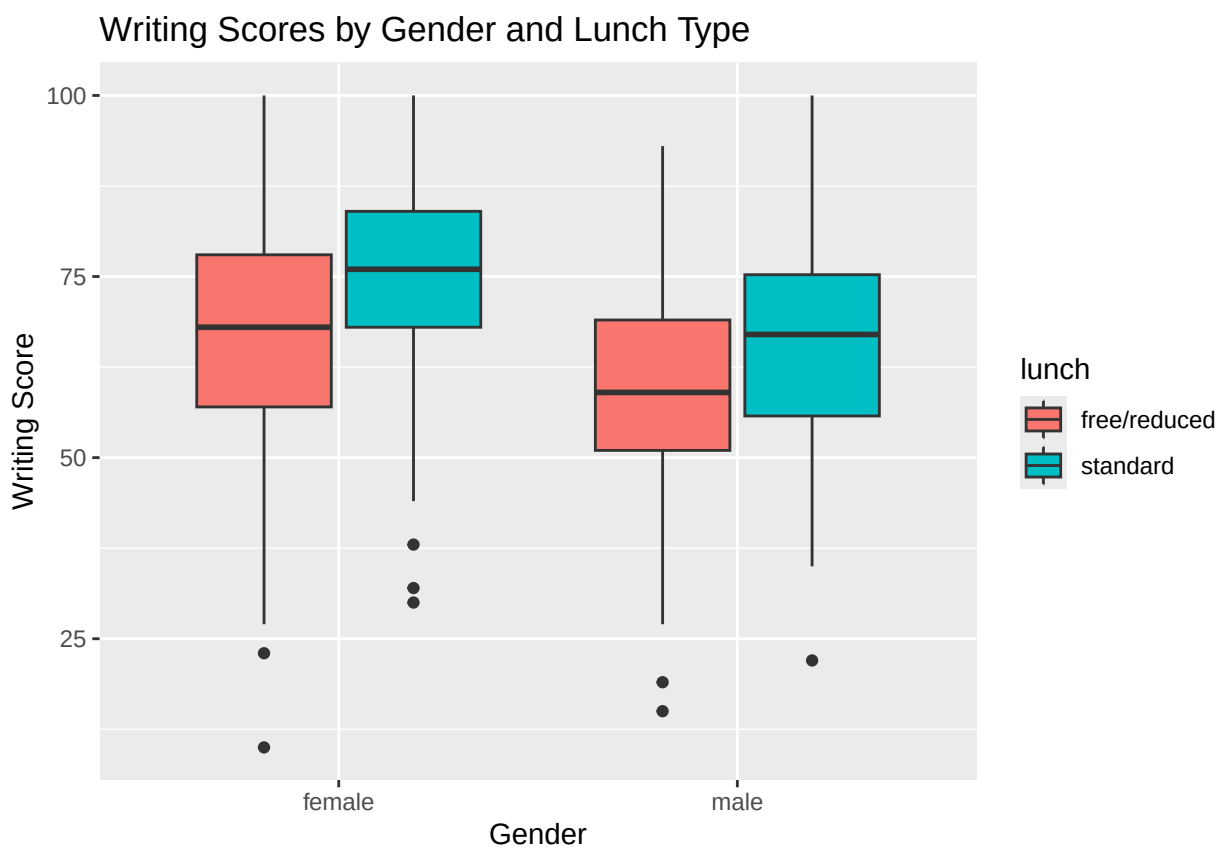

```r
# Identify median scores
df %>%
  group_by(parental_level_of_education) %>%
  summarise(median_math = median(math_score)) %>%
  arrange(desc(median_math))
```

```
## # A tibble: 6 x 2
```

```
##   parental_level_of_education median_math
##   <chr>                             <dbl>
## 1 master's degree                    73
## 2 bachelor's degree                  68
## 3 some college                       67.5
## 4 associate's degree                 67
## 5 some high school                   65
## 6 high school                        63
```

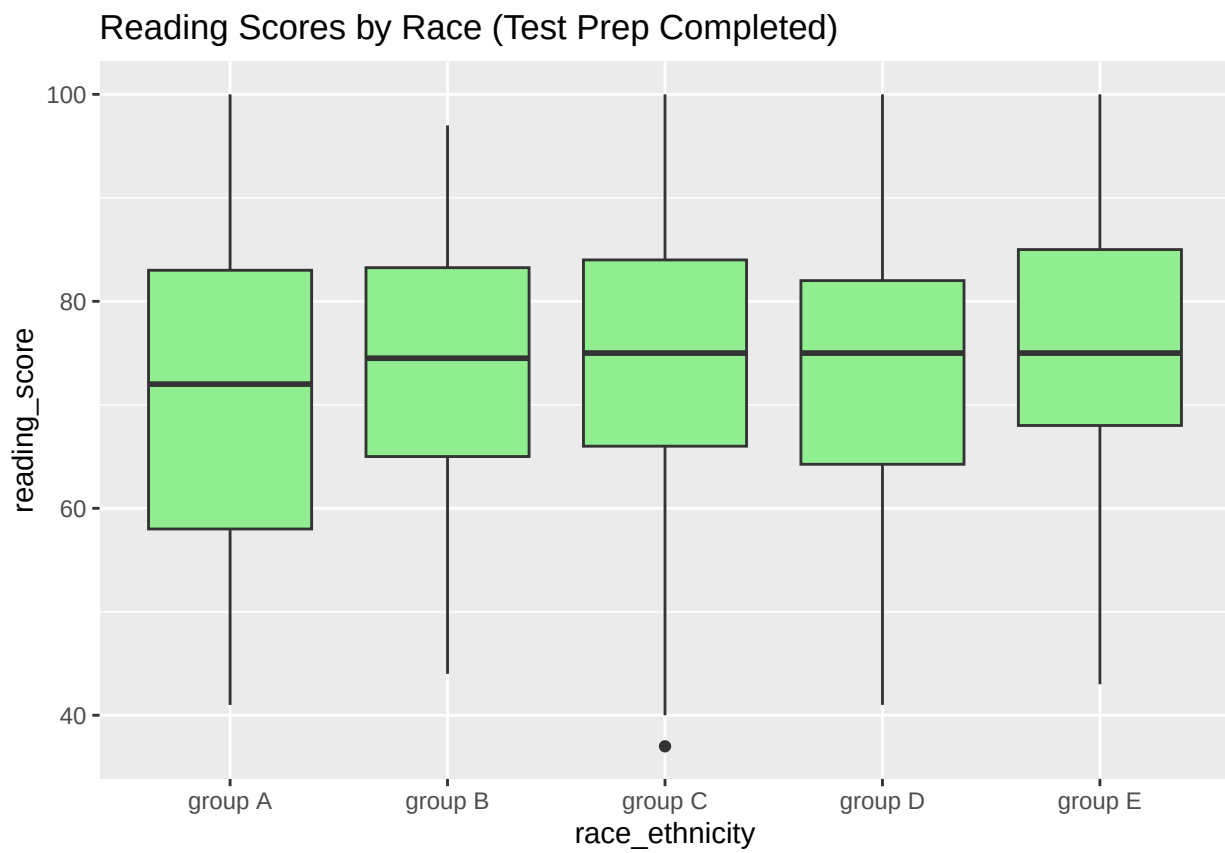
##QUESTION 2.WRITING SCORES BY GENDER AND LUNCH TYPE

```
ggplot(df, aes(x = gender, y = writing_score, fill = lunch)) +
  geom_boxplot() +
  labs(title = "Writing Scores by Gender and Lunch Type",
       x = "Gender", y = "Writing Score")
```



##QUESTION 3.READING SCORES (COMPLETED PREP)BY RAC/ETHNICITY

```
prep_completed <- df %>% filter(test_preparation_course == "completed")

ggplot(prep_completed, aes(x = race_ethnicity, y = reading_score)) +
  geom_boxplot(fill = "lightgreen") +
  labs(title = "Reading Scores by Race (Test Prep Completed)")
```

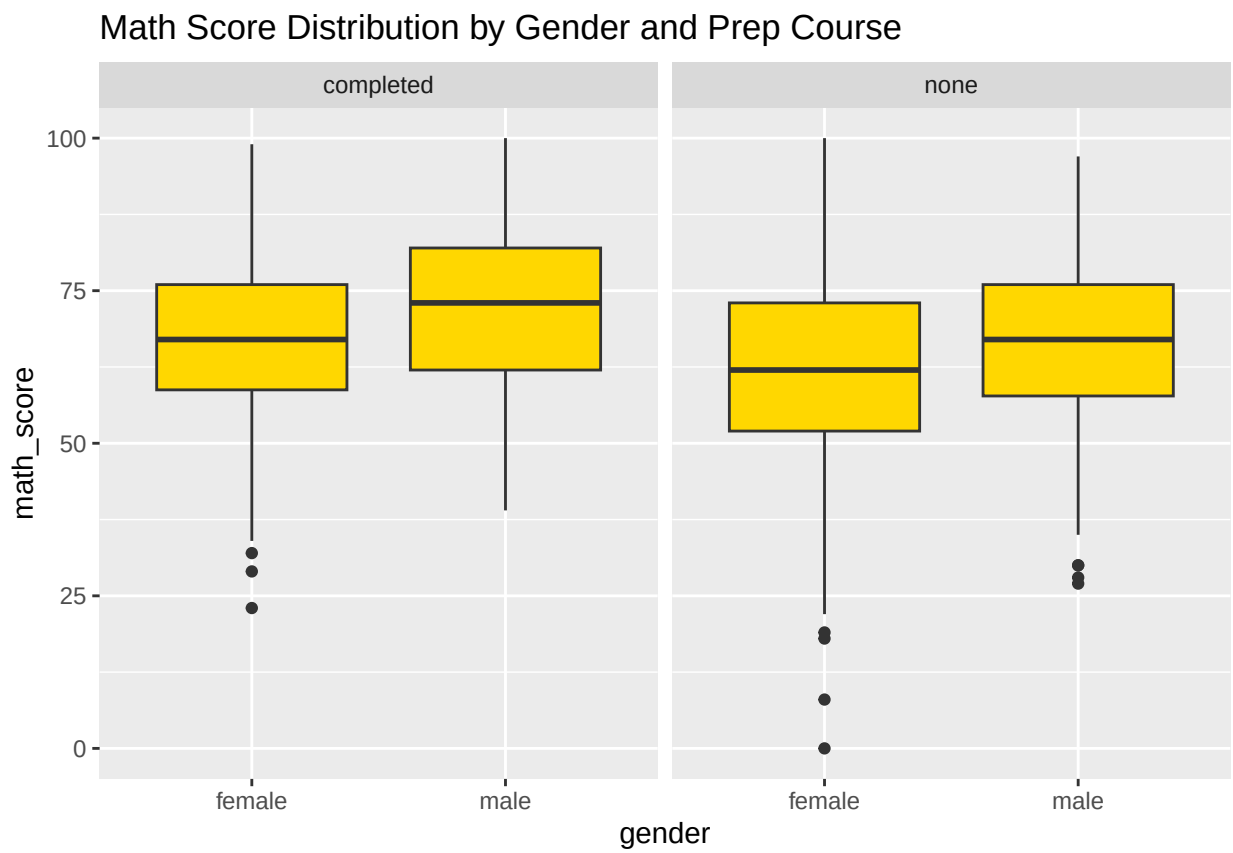



```
# Calculate IQR to identify largest variability
prep_completed %>%
  group_by(race_ethnicity) %>%
  summarise(IQR_reading = IQR(reading_score)) %>%
  arrange(desc(IQR_reading))
```

```
## # A tibble: 5 x 2
##   race_ethnicity IQR_reading
##   <chr>                <dbl>
## 1 group A               25
## 2 group B               18.2
## 3 group C               18
## 4 group D               17.8
## 5 group E               17
```

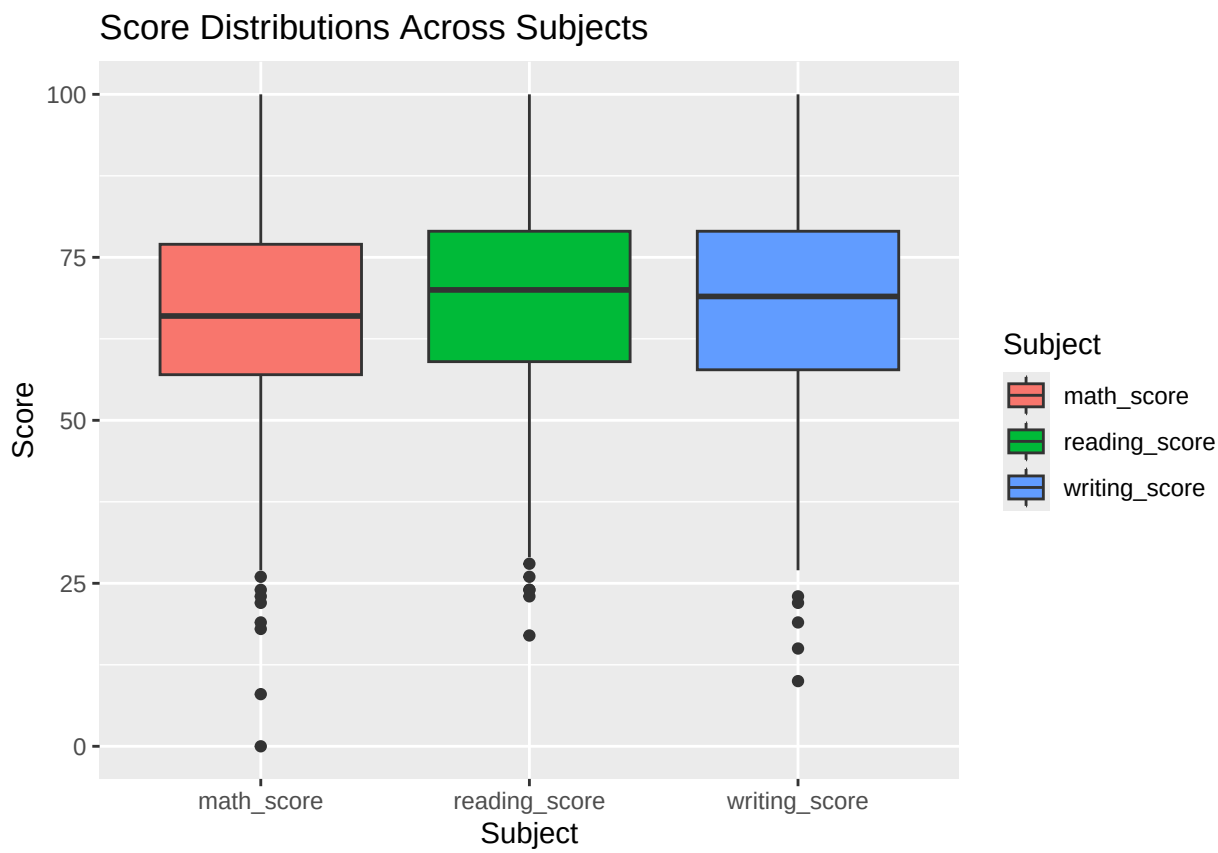
##QUESTION 4.FACETED MATHS SCORES BY PREPARATION STATUS

```
ggplot(df, aes(x = gender, y = math_score)) +
  geom_boxplot(fill = "gold") +
  facet_wrap(~test_preparation_course) +
  labs(title = "Math Score Distribution by Gender and Prep Course")
```



##QUESTION 5.SUBJECT COMPARISON

```
df_long <- df %>%
  pivot_longer(cols = ends_with("score"), names_to = "Subject", values_to = "Score")

ggplot(df_long, aes(x = Subject, y = Score, fill = Subject)) +
  geom_boxplot() +
  labs(title = "Score Distributions Across Subjects")
```

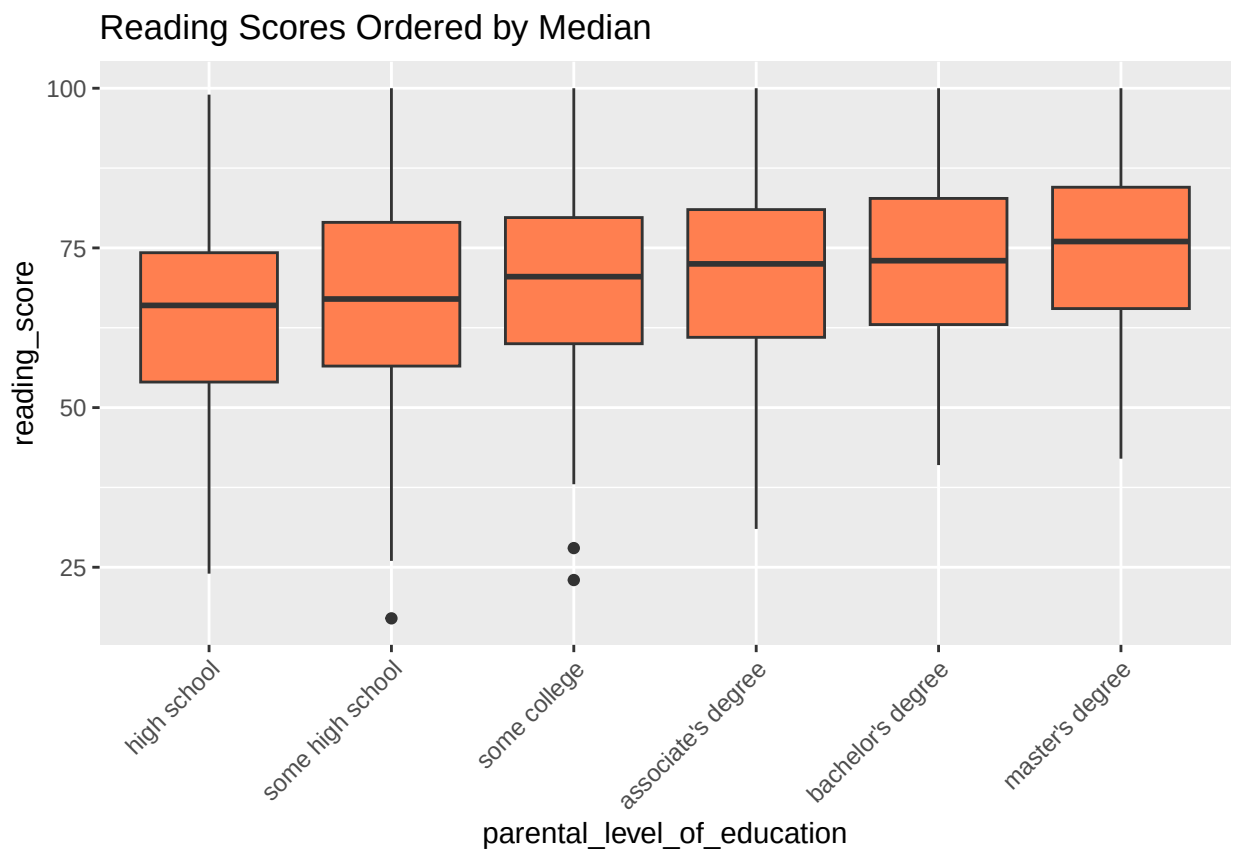



```
# Check standard deviation for variability
df_long %>%
  group_by(Subject) %>%
  summarise(SD_Score = sd(Score))
```

```
## # A tibble: 3 x 2
##   Subject       SD_Score
##   <chr>            <dbl>
## 1 math_score        15.2
## 2 reading_score     14.6
## 3 writing_score     15.2
```

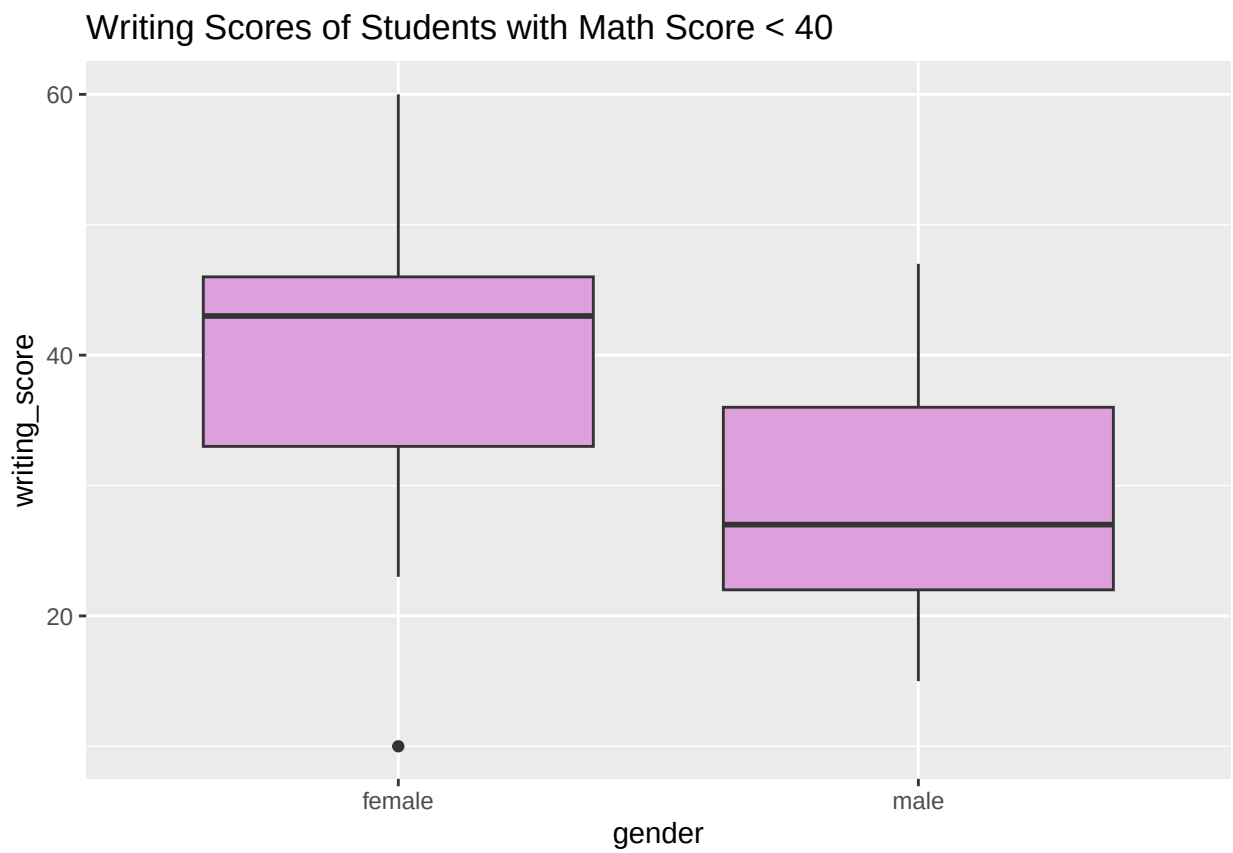
##QUESTION 6.REDING SCORES ORDERED BY MEDIAN

```
# Order factor levels based on median reading score
df_ordered <- df %>%
  mutate(parental_level_of_education = fct_reorder(parental_level_of_education, reading_score, .fun = me

ggplot(df_ordered, aes(x = parental_level_of_education, y = reading_score)) +
  geom_boxplot(fill = "coral") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Reading Scores Ordered by Median")
```



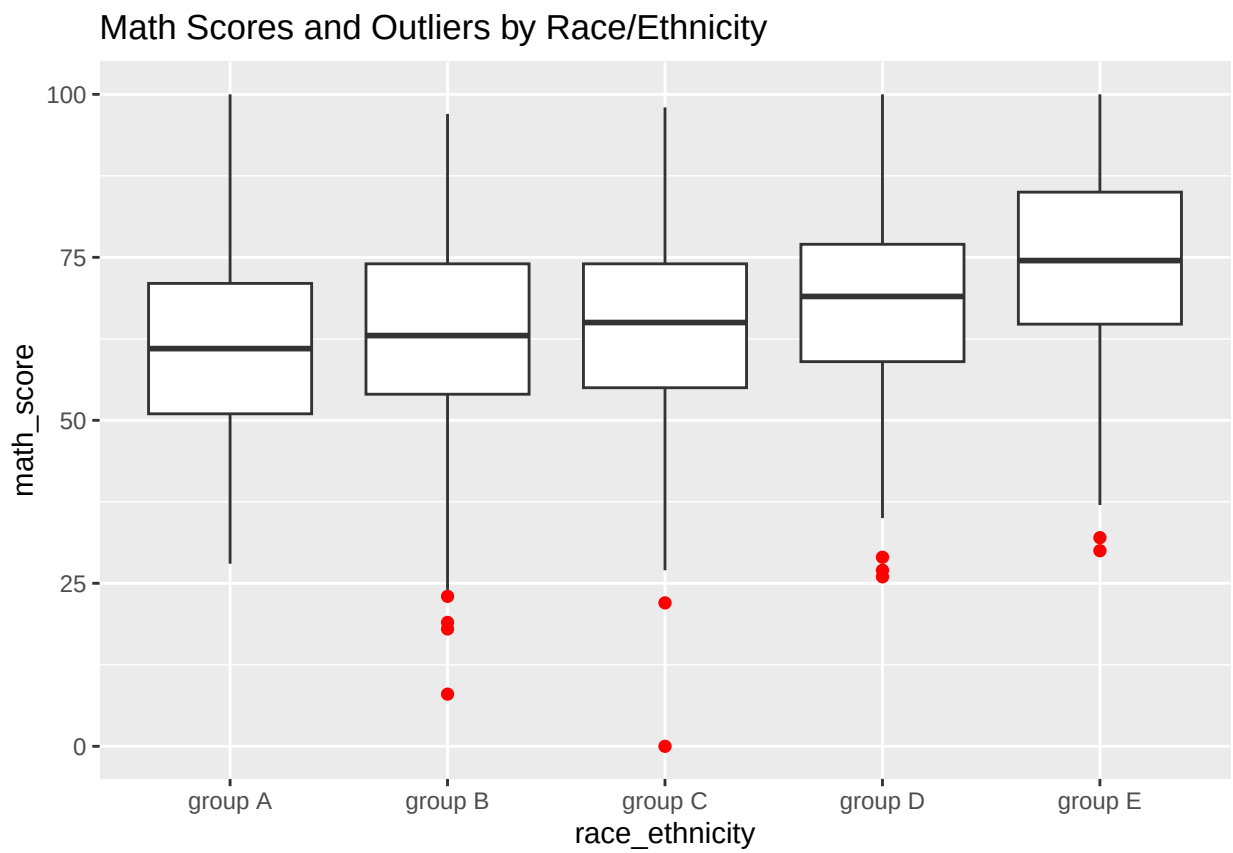
##QUESTION 7.POOR MATHS PERFOMANCE(<-40) AND WRITING

```
low_math <- df %>% filter(math_score < 40)

ggplot(low_math, aes(x = gender, y = writing_score)) +
  geom_boxplot(fill = "plum") +
  labs(title = "Writing Scores of Students with Math Score < 40")
```



##QUESTION 8.MATH OUTLIERS BY RACE/ETHNICITY

```
ggplot(df, aes(x = race_ethnicity, y = math_score)) +
  geom_boxplot(outlier.colour = "red", outlier.shape = 16, outlier.size = 2) +
  labs(title = "Math Scores and Outliers by Race/Ethnicity")
```

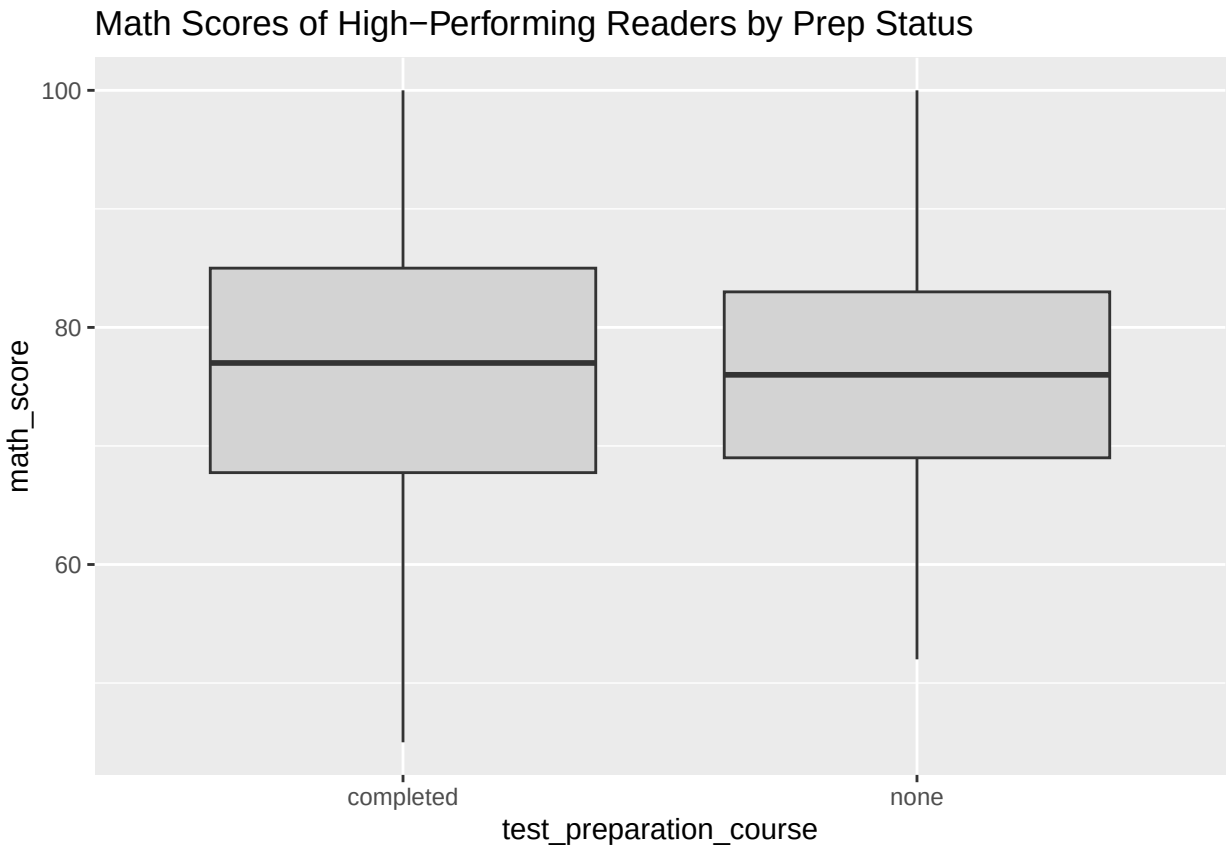


##QUESTION 9.FACETED WRITING SCORES BY LUNCH TYPE

```
ggplot(df, aes(x = gender, y = writing_score)) +
  geom_boxplot(fill = "lightblue") +
  facet_grid(. ~ lunch) +
  labs(title = "Writing Scores Faceted by Lunch Type")
```

##QUESTION 10.MATHS SCORES FOR HIGH PERFOMING READERS

```
med_reading <- median(df$reading_score)
high_readers <- df %>% filter(reading_score > med_reading)

ggplot(high_readers, aes(x = test_preparation_course, y = math_score)) +
  geom_boxplot(fill = "lightgrey") +
  labs(title = "Math Scores of High-Performing Readers by Prep Status")
```


Math Scores of High−Performing Readers by Prep Status
100
80
60
math_score
completed
none
test_preparation_course